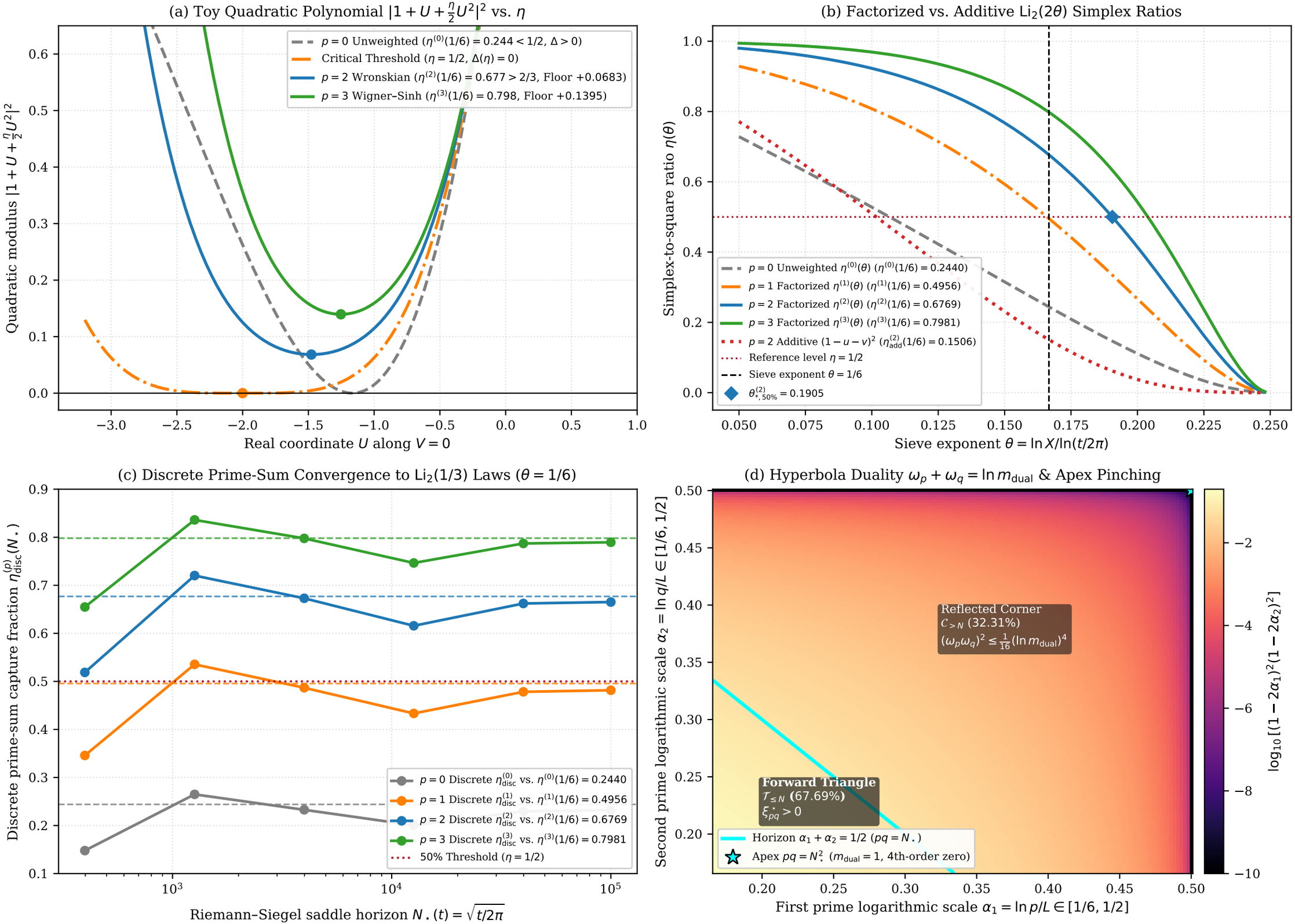

(a) Toy Quadratic Polynomial $|1 + U + \frac{\eta}{2}U^2|^2$ vs. η
$p = 0$ Unweighted ($\eta^{(0)}(1/6) = 0.244 < 1/2$, $\Delta > 0$)
Critical Threshold ($\eta = 1/2$, $\Delta(\eta) = 0$)
$p = 2$ Wronskian ($\eta^{(2)}(1/6) = 0.677 > 2/3$, Floor +0.0683)
$p = 3$ Wigner–Sinh ($\eta^{(3)}(1/6) = 0.798$, Floor +0.1395)
Quadratic modulus $|1 + U + \frac{\eta}{2}U^2|^2$
Real coordinate U along $V = 0$
(b) Factorized vs. Additive $\mathrm{Li}_2(2\theta)$ Simplex Ratios
$p = 0$ Unweighted $\eta^{(0)}(\theta)$ ($\eta^{(0)}(1/6) = 0.2440$)
$p = 1$ Factorized $\eta^{(1)}(\theta)$ ($\eta^{(1)}(1/6) = 0.4956$)
$p = 2$ Factorized $\eta^{(2)}(\theta)$ ($\eta^{(2)}(1/6) = 0.6769$)
$p = 3$ Factorized $\eta^{(3)}(\theta)$ ($\eta^{(3)}(1/6) = 0.7981$)
$p = 2$ Additive $(1 - u - v)^2$ ($\eta^{(2)}_{\mathrm{add}}(1/6) = 0.1506$)
Reference level $\eta = 1/2$
Sieve exponent $\theta = 1/6$
$\theta^{(2)}_{*,50\%} = 0.1905$
Simplex-to-square ratio $\eta(\theta)$
Sieve exponent $\theta = \ln X / \ln(t/2\pi)$
(c) Discrete Prime-Sum Convergence to $\mathrm{Li}_2(1/3)$ Laws ($\theta = 1/6$)
$p = 0$ Discrete $\eta^{(0)}_{\mathrm{disc}}$ vs. $\eta^{(0)}(1/6) = 0.2440$
$p = 1$ Discrete $\eta^{(1)}_{\mathrm{disc}}$ vs. $\eta^{(1)}(1/6) = 0.4956$
$p = 2$ Discrete $\eta^{(2)}_{\mathrm{disc}}$ vs. $\eta^{(2)}(1/6) = 0.6769$
$p = 3$ Discrete $\eta^{(3)}_{\mathrm{disc}}$ vs. $\eta^{(3)}(1/6) = 0.7981$
50% Threshold ($\eta = 1/2$)
Discrete prime-sum capture fraction $\eta^{(p)}_{\mathrm{disc}}(N_\star)$
Riemann–Siegel saddle horizon $N_\star(t) = \sqrt{t/2\pi}$
(d) Hyperbola Duality $\omega_p + \omega_q = \ln m_{\mathrm{dual}}$ & Apex Pinching
Reflected Corner
$\mathcal{C}_{>N}$ (32.31%)
$(\omega_p \omega_q)^2 \le \frac{1}{16}(\ln m_{\mathrm{dual}})^4$
Forward Triangle
$\mathcal{T}_{\le N}$ (67.69%)
$\xi^\star_{pq} > 0$
Horizon $\alpha_1 + \alpha_2 = 1/2$ ($pq = N_\star$)
Apex $pq = N_\star^2$ ($m_{\mathrm{dual}} = 1$, 4th-order zero)
Second prime logarithmic scale $\alpha_2 = \ln q / L \in [1/6, 1/2]$
First prime logarithmic scale $\alpha_1 = \ln p / L \in [1/6, 1/2]$
$\log_{10}[(1 - 2\alpha_1)^2(1 - 2\alpha_2)^2]$